Pedro Núñez

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Work Experience

Polytechnic University of Madrid, Profesor Ayudante Doctor — Madrid, Spain
- In the School of Computer Systems Engineering (ETSISI). — Sept 2026 – present

National Taiwan University, Postdoctoral Researcher — Taipei, Taiwan
- In the group of Jungkai Alfred Chen. — Sept 2023 – July 2026
- Research focused on birational geometry and derived categories of coherent sheaves.
- Participated in and organized various seminars and reading groups.

Participation in Research Projects

Varieties of general type (NSTC114-2639-M-002-009-ASP) — Taipei, Taiwan
Funded by National Science and Technology Council of Taiwan. PI: Jungkai Alfred Chen. — Sept 2023 – July 2026

GRK 1821: Cohomological Methods in Geometry — Freiburg, Germany
Funded by Deutsche Forschungsgemeinschaft (DFG). — Sept 2019 – Aug 2021

Publications

Derived categories of threefolds near the Noether line — 2026
Jungkai Alfred Chen, *Pedro Núñez*.
Preprint (arxiv.org/abs/2610.29912).

Extension of adapted differentials on klt orbifolds — 2024
Pedro Núñez.
Preprint (arxiv.org/abs/2411.17268).

The Albanese morphism for hyperelliptic varieties — 2025
Pieter Belmans, Andreas Demleitner, *Pedro Núñez*.
Indagationes Mathematicae (10.1016/j.indag.2025.10.004).

Adapted differentials as a qfh-sheaf — 2023
Pedro Núñez.
Mathematische Zeitschrift 304, 68 (10.1007/s00209-023-03330-6).

A note on semiorthogonal decompositions for Fano fibrations — 2022
Pedro Núñez.
Geometriae Dedicata 216, 20 (10.1007/s10711-022-00680-z).

Teaching Experience

Polytechnic University of Madrid, Lecturer — Madrid, Spain
- Lecturer for *Logic and Discrete Mathematics* for first-year computer engineering majors. — Sept 2026 – present
- Taught in Spanish.

National Taiwan University, Lecturer — Taipei, Taiwan
- Lecturer for *Calculus 1*, *Calculus 2*, *Calculus 3*, and *Calculus 4* for first-year economics majors. — Sept 2024 – July 2025
- Taught theoretical classes, coordinated the course and tutors, and created exam problems.
- Taught in English.

University of Freiburg, Teaching Assistant — Freiburg, Germany
Oct 2021 – July 2023

- Teaching Assistant for *Algebra and Number Theory*, *Elementary Geometry* (twice), and *Riemann Surfaces*.
- Handled question sessions for *Linear Algebra II*.
- Created weekly exercise sheets, coordinated practice classes, offered tutoring, and discussed solutions.
- Taught in German.

University of Bonn, Tutor — Bonn, Germany
Apr 2018 – July 2019

- Tutor for *Linear Algebra for Informatics and Teachers*, and *Introduction to Algebra*.
- Graded weekly homework and discussed solutions in exercise classes.
- Taught in German.

Education

PhD **University of Freiburg**, PhD in Mathematics — Freiburg, Germany
Sept 2019 – July 2023

- Thesis: Categorical aspects of Campana orbifolds, supervised by Stefan Kebekus.
- PhD title certified in Spain by the Autonomous University of Madrid.
- Grade: magna cum laude.

MSc **University of Bonn**, Master in Mathematics — Bonn, Germany
Oct 2017 – July 2019

- Thesis: Derived categories of Fano fibrations.
- Grade: 1.2 in the German system (approx. 9.5/10 in the Spanish system).

BSc **Complutense University of Madrid**, Bachelor in Mathematics — Madrid, Spain
Sept 2013 – Aug 2017

- Thesis: Blow-ups in algebraic geometry.
- Grade: 8.34/10 in the Spanish system.
- Erasmus exchange at LMU Munich during the course 2016/2017.

Languages

Spanish: Native speaker.

English: C1 level (IELTS with score 8).

German: C1 level (Goethe-Zertifikat C1).

Chinese: B1 level (HSK3 + HSKK Basic).

Invited Talks

Indecomposability of derived categories of hyperelliptic varieties — Taipei, Taiwan
2026 Higher Dimensional Algebraic Geometry. — 17 Apr 2026

Campana orbifolds and their morphisms — Chiayi, Taiwan
61st Annual Meeting of the Taiwan Mathematical Society. — 22 Jan 2026

Campana orbifolds: between projective and logarithmic geometry — Taipei, Taiwan
Taipei Postdoc Seminar. — 5 Nov 2025

Adapted differentials on Campana orbifolds — Seoul, South Korea
KIAS Algebraic Geometry Seminar. — 31 Oct 2025

Derived categories and birational geometry — Taoyuan, Taiwan
Colloquium talk at National Central University. — 23 Oct 2025

Derived categories of threefolds on the Noether line — Shanghai, China
SJTU Algebraic Geometry Seminar. — 10 Oct 2025

Derived categories of threefolds on the Noether line — Shanghai, China
Fudan University Algebraic Geometry Seminar. — 8 Oct 2025

Derived categories and birational geometry — Madrid, Spain
Mathematics Seminar at CUNEF University. — 22 Sep 2025

Derived categories of hyperelliptic varieties — Tainan, Taiwan
Algebraic Geometry meets Machine Learning. — 29 Apr 2025

Indecomposability of derived categories of threefolds on the Noether line — Penghu, Taiwan
Younger Generation of Algebraic Geometry 2025. — 19 Apr 2025

On the Albanese morphism of hyperelliptic varieties — Taipei, Taiwan
NCTS One-Day Workshop on Arithmetic and Algebraic Geometry. — 20 Feb 2025

Indecomposability of derived categories of threefolds on the Noether line — Tainan, Taiwan
NCKU Algebraic Geometry Seminar. — 18 Apr 2024

Indecomposability of derived categories of hyperelliptic manifolds — Taipei, Taiwan
NCTS Seminar in Algebraic Geometry. — 20 Oct 2023

Adapted differentials on Campana orbifolds — Esch-sur-Alzette, Luxembourg
Luxembourg Algebraic Geometry Seminar. — 22 Mar 2023

Adapted differentials on Campana orbifolds — Taipei, Taiwan
NCTS Seminar in Algebraic Geometry (online). — 6 Feb 2023

Categorías derivadas y geometría birracional — Madrid, Spain
Red de Doctorandos en Matemáticas UCM (online). — 22 Mar 2022

Contributed Talks

Indecomposability of derived categories of hyperelliptic varieties — Fort Collins, CO, USA
2025 Summer Research Institute in Algebraic Geometry. — 22 Jul 2025

Adapted differentials as a qfh-sheaf — Seville, Spain
9th European Congress of Mathematics. — 15 Jul 2024

Semiorthogonal decompositions on Fano fibrations — Orsay, France
Géométrie Algébrique en Liberté XXIX. — 24 May 2022

Derived categories of Fano varieties and Mori fibre spaces — Madrid, Spain
2nd BYMAT Conference. — 21 May 2019

Academic Visits

Academic visit at Fudan University — Shanghai, China
Shanghai Center for Mathematical Sciences. Hosted by Meng Chen. — 28 Sep – 11 Oct 2025

Other Academic Activities

Organization of the international conference Tour de Formosa — Taiwan's East Coast
Organized jointly with Flora Poon. — 12 – 16 Jan 2026
sites.google.com/view/tour-de-formosa

Organization of the NCTS Seminar in Algebraic Geometry — Taipei, Taiwan
Invited external speakers to talk about their research. — Summer Term 2025

Organization of a seminar on deformation theory — Taipei, Taiwan
Organized jointly with Flora Poon. — Summer Term 2025

Organization of a seminar on Hodge theory — Taipei and Xitou, Taiwan
Organized jointly with Hsueh-Yung Lin. — Summer 2025

Organization of a seminar on enumerative geometry — Freiburg, Germany
Notes for the seminar are available on my homepage. — Winter Term 2020/2021

Organization of a reading group on Grothendieck's EGA — Bonn, Germany
Informal reading group with various other students. — Winter Term 2017/2018

Scholarships & Funding received

2025 Summer Research Institute in Algebraic Geometry — Fort Collins, CO, USA
Funded by the NCTS and NSTC to attend the conference. — 12 – 26 Jul 2025

Geometric and Arithmetic Frontiers of Orbifolds II — Strasbourg, France
Funded by IRMA to attend their workshop. — 5 – 9 Jun 2023

Oberwolfach Leibniz Graduate Students (OWLG) — Oberwolfach, Germany
Funded by MFO to attend the Komplexe Analysis workshop. — 9 – 14 Apr 2023

2nd BYMAT Conference — Madrid, Spain
Funded by ICMAT to attend their conference. — 20 – 24 May 2019

Geometry Summer School — Utrecht, Netherlands
Funded by Utrecht University to attend their summer school. — 14 – 25 Aug 2017

Erasmus Scholarship — Munich, Germany
Funded by the European Union to study as an exchange student. — Sep 2016 – Jul 2017

Dynamical Systems Summer School — Lisbon, Portugal
Funded by the Fundação Calouste Gulbenkian to attend their summer school. — 6 – 10 Jul 2015

Conferences, Workshops and Schools Attended

Mini-Workshop: Algebraic Geometry in Xitou — Xitou, Taiwan
National Taiwan University / National Science and Technology Council. — 19 – 21 Jul 2026
sites.google.com/view/mini-workshopalgebraicgeometry/%E9%A6%96%E9%A0%81

10th Pacific Rim Conference on Mathematics — Madrid, Spain
Universidad Complutense de Madrid. — 15 – 19 Jun 2026
sites.google.com/view/2026pacificrim/home

Topological and Geometric Deep Learning: Theory, Methods, and Applications — Taipei, Taiwan
Academia Sinica. — 11 – 14 May 2026
ines.topology.rocks/tgdl-2026

NCTS Minicourse - Topics on Derived Categories of Coherent Sheaves — Taipei, Taiwan
National Center for Theoretical Sciences. Lecturer: Yujiro Kawamata. — 22, 27, 29 Apr 2026
ncts.ntu.edu.tw/events_3_detail.php?nid=395

2026 Higher Dimensional Algebraic Geometry — Taipei, Taiwan
National Taiwan University. — 13 – 17 Apr 2026
sites.google.com/ncts.ntu.edu.tw/2026hdag

2026 Higher Dimensional Algebraic Geometry - Minicourses — Taipei, Taiwan
National Taiwan University. — 4 – 10 Apr 2026
sites.google.com/ncts.ntu.edu.tw/2026hdag-minicourses

Meet with Prof. Chenyang Xu — Taipei, Taiwan
National Center for Theoretical Sciences. — 2 Apr 2026
ncts.ntu.edu.tw/events_2_detail.php?nid=552

61st Annual Meeting of the Taiwan Mathematical Society — Chiayi, Taiwan
National Chung Cheng University. — 21 – 22 Jan 2026
2026tms-en.tms.org.tw

Tour de Formosa — Taiwan's East Coast
National Taiwan University / National Science and Technology Council. — 12 – 16 Jan 2026
sites.google.com/view/tour-de-formosa

NCTS Minicourse on Gizatullin's problem — Taipei, Taiwan
National Center for Theoretical Sciences. Lecturer: Keiji Oguiso. — 24, 26 Nov, 1 Dec 2025
ncts.ntu.edu.tw/events_3_detail.php?nid=384

2025 Algebraic Geometry in East Asia — Tainan, Taiwan
National Cheng Kung University. — 10 – 14 Nov 2025
sites.google.com/ncts.ntu.edu.tw/2025-algebraic-geometry-in/2025-agea

2025 Summer Research Institute in Algebraic Geometry — Fort Collins, CO, USA
Colorado State University. — 14 Jul – 1 Aug 2025
sites.google.com/view/2025summerinstitute/home

NCTS Minicourse on Deformation Theory — Taipei, Taiwan
National Center for Theoretical Sciences. Lecturer: Yujiro Kawamata. — 2, 9 May 2025
ncts.ntu.edu.tw/events_3_detail.php?nid=369

Algebraic Geometry meets Machine Learning — Tainan, Taiwan
National Cheng Kung University. — 27 – 29 Apr 2025
sites.google.com/view/cjlai-ncku/conferenceworkshop/ag-meets-ml-ncku-2025-4

Younger Generation of Algebraic Geometry 2025 — Penghu, Taiwan
National Taiwan University / National Science and Technology Council. — 17 – 19 Apr 2025
sites.google.com/view/laboratoryofbirationalgeometry/%E9%A6%96%E9%A0%81

NCTS One-Day Workshop on Arithmetic and Algebraic Geometry — Taipei, Taiwan
National Taiwan University. — 20 Feb 2025
ncts.ntu.edu.tw/events_2_detail.php?nid=492

60th Annual Meeting of the Taiwan Mathematical Society — Taoyuan, Taiwan
National Central University. — 14 – 15 Jan 2025
2025tms.tms.org.tw

Introduction to Nevanlinna Theory, Hyperbolicity and Arithmetic Dynamics — Taipei, Taiwan
Academia Sinica. — 2 – 3 Jan 2025
www.math.sinica.edu.tw/workshop/202501SCV/index.html

Algebraic Geometry in Taipei — Taipei, Taiwan
National Taiwan University. — 2 – 6 Dec 2024
sites.google.com/g.ntu.edu.tw/ag-Taipei/mpbg-2023

Mini-Workshop: Young Perspectives in Algebraic Geometry — Taipei, Taiwan
National Taiwan University. — 25 – 26 Nov 2024
sites.google.com/view/mini-workshop-young-perspectiv/%E9%A6%96%E9%A0%81

Workshop in Hodge Theory — Xitou, Taiwan
National Taiwan University / National Science and Technology Council. — 21 – 23 Aug 2024
sites.google.com/view/laboratoryofbirationalgeometry/%E9%A6%96%E9%A0%81

9th European Congress of Mathematics — Seville, Spain
Universidad de Sevilla. — 15 – 19 Jul 2024
www.ecm2024sevilla.com

Transcendental aspects of algebraic geometry — Cetraro, Italy
Grand Hotel San Michele. — 1 – 5 Jul 2024
www2.mathematik.hu-berlin.de/~klingleb/Cetraro24/main.html

NCTS Minicourse on Non-Commutative Geometry — Taipei, Taiwan
National Center for Theoretical Sciences. Lecturer: Yujiro Kawamata. — 17, 24, 29 Apr 2024
ncts.ntu.edu.tw/events_3_detail.php?nid=334

Mini-Arbeitsgemeinschaft 2024: Severi-Brauer varieties — Hualien, Taiwan
National Taiwan University / National Science and Technology Council. — 26 – 29 Jan 2024
sites.google.com/view/laboratoryofbirationalgeometry/%E9%A6%96%E9%A0%81

59th Annual Meeting of the Taiwan Mathematical Society — Taipei, Taiwan
National Chengchi University. — 22 – 23 Jan 2024
2024tms.tms.org.tw

NCTS-Kyoto Mathematics Symposium — Taipei, Taiwan
National Center for Theoretical Sciences. — 9 Dec 2023
ncts.ntu.edu.tw/events_2_detail.php?nid=427

Birational Geometry and Algebraic Dynamics — Tokyo, Japan
University of Tokyo. — 27 Nov – 1 Dec 2023
www.ms.u-tokyo.ac.jp/~stakagi/workshop/oguiso/index.html

Algebraic Geometry in East Asia — Seoul, South Korea
Korea Institute For Advanced Study. — 6 – 10 Nov 2023
events.kias.re.kr/h/AGEA23

Geometric and Arithmetic Frontiers of Orbifolds II — Strasbourg, France
Institut de recherche mathématique avancée. — 5 – 9 Jun 2023
cirget.uqam.ca/Orbifolds3/en

Differential and Algebraic methods in Kähler spaces — Oberwolfach, Germany
Mathematisches Forschungsinstitut Oberwolfach. — 9 – 14 Apr 2023
www.mfo.de/occasion/2315/www_view

Crash Course Condensed Mathematics — Freiburg, Germany
University of Freiburg. — 13 – 17 Mar 2023
math.commelin.net/2023/condensed

Abschlusskolloquium of GK "Cohomological Methods in Geometry" — Freiburg, Germany
University of Freiburg. — 14 – 15 Jul 2022
freidok.uni-freiburg.de/proj/8331

Birational Geometry conference — Lausanne, Switzerland
École Polytechnique Fédérale de Lausanne. — 13 – 17 Jun 2022
www.epfl.ch/labs/cag/birational-geometry-conference

Géométrie Algébrique en Liberté XXIX — Orsay, France
Laboratoire de Mathématiques d'Orsay. — 23 – 27 May 2022
lmv.math.cnrs.fr/geometrie-algebrique-en-liberte-gael-xxix

Workshop Higher Algebra — Darmstadt, Germany
Technical University of Darmstadt. — 16 – 18 Mar 2022
crc326gaus.de/workshop-higher-algebra

New Techniques in Resolution of Singularities (online) — Oberwolfach, Germany
Mathematisches Forschungsinstitut Oberwolfach. — 24 – 30 Oct 2021
www.mfo.de/www/activity/2143b

K-theory, algebraic cycles and motivic homotopy theory — Cambridge, UK
Isaac Newton Institute for Mathematical Sciences. — 13 – 17 Jan 2020
www.newton.ac.uk/event/kahw01

o-Minimal Structures in Algebraic Geometry Summer School — Freiburg, Germany
University of Freiburg. — 9 – 13 Sep 2019
www.math.ens.psl.eu/~odegaay/Freiburg.pdf

Moduli and Stability conditions Summer School — Hannover, Germany
University of Hannover. — 29 Jul – 2 Aug 2019
sites.google.com/site/moduli2019hannover/home

2nd BYMAT Conference — Madrid, Spain
Instituto de Ciencias Matemáticas. — 20 – 24 May 2019
www.icmat.es/congresos/2019/BYMAT/abstracts.pdf

Geometry Summer School — Utrecht, Netherlands
Utrecht University. — 14 – 25 Aug 2017
www.utrechtsummerschool.nl/courses/science/geometry

Partial Differential Equations Summer School — Lisbon, Portugal
Fundação Calouste Gulbenkian. — 4 – 8 Jul 2016
www.math.tecnico.ulisboa.pt/~ggranja/Talentos/school2016

Dynamical Systems Summer School — Lisbon, Portugal
Fundação Calouste Gulbenkian. — 6 – 10 Jul 2015
www.math.tecnico.ulisboa.pt/~ggranja/Talentos/school2015

Algebraic Geometry Summer School — Lisbon, Portugal
Fundação Calouste Gulbenkian. — 14 – 18 Jul 2014
www.math.tecnico.ulisboa.pt/~ggranja/Talentos/school2014

Additional Formation

Internship at the WSO-UV space telescope project — Madrid, Spain
Time-series analysis from ultraviolet spectroscopy measurements using Matlab. — 1 – 31 Jul 2015

International Astronomical Youth Camp 2014 — Weyer, Austria
Observational astronomy and climate modeling using Matlab. — 20 Jul – 9 Aug 2014

Android application development I, II and III Madrid, Spain
Courses taken at TEIDE Centros de Formación. Jan – Jun 2013